

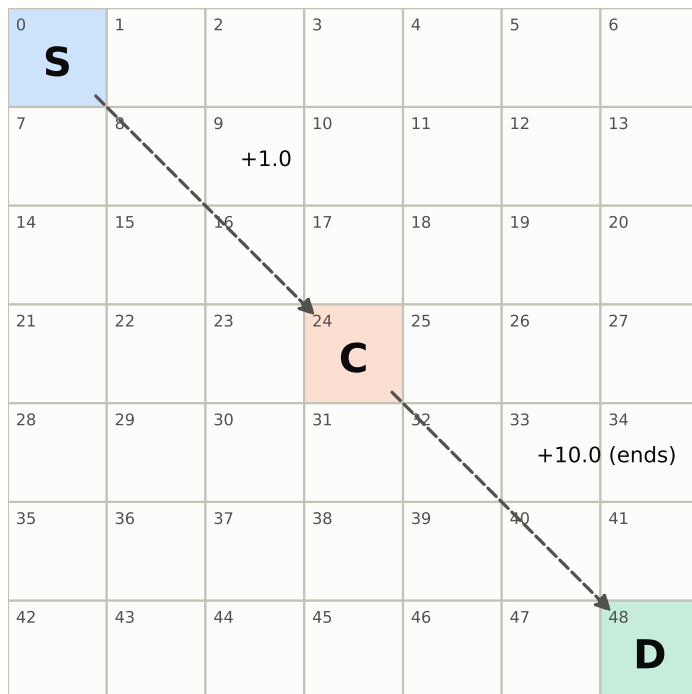
CS272 PA2: Courier Route and SARSA(λ)

Repository: <https://github.com/Kushagrabainisla/cs272-pa2-gym>

1. The environment

Courier Route is a 7×7 warehouse floor. The courier starts at the depot S in the upper-left corner, must walk to the package checkpoint C in the middle of the floor and pick the package up, and then carry it to the delivery dock D in the lower-right corner; the episode ends on delivery. There are 4 actions (up, right, down, left) and 98 states: 2 stages (package not yet / already collected) $\times$ 49 cells, encoded as $\text{state} = \text{stage} \cdot 49 + \text{row} \cdot 7 + \text{column}$. Every step costs -0.02 , the first arrival on C pays $+1.0$, and delivering pays $+10.0$. The floor is slippery: with probability 0.85 a move goes as requested, and with probability 0.15 it slips to one of the two perpendicular directions (0.075 each); walls block. Episodes are truncated after 100 steps. The pickup and the delivery are each 6 moves from the previous stop, so the reward that matters sits about 12 steps after the first decision that earns it.

State diagram (stage 0 ids; add 49 after the pickup)

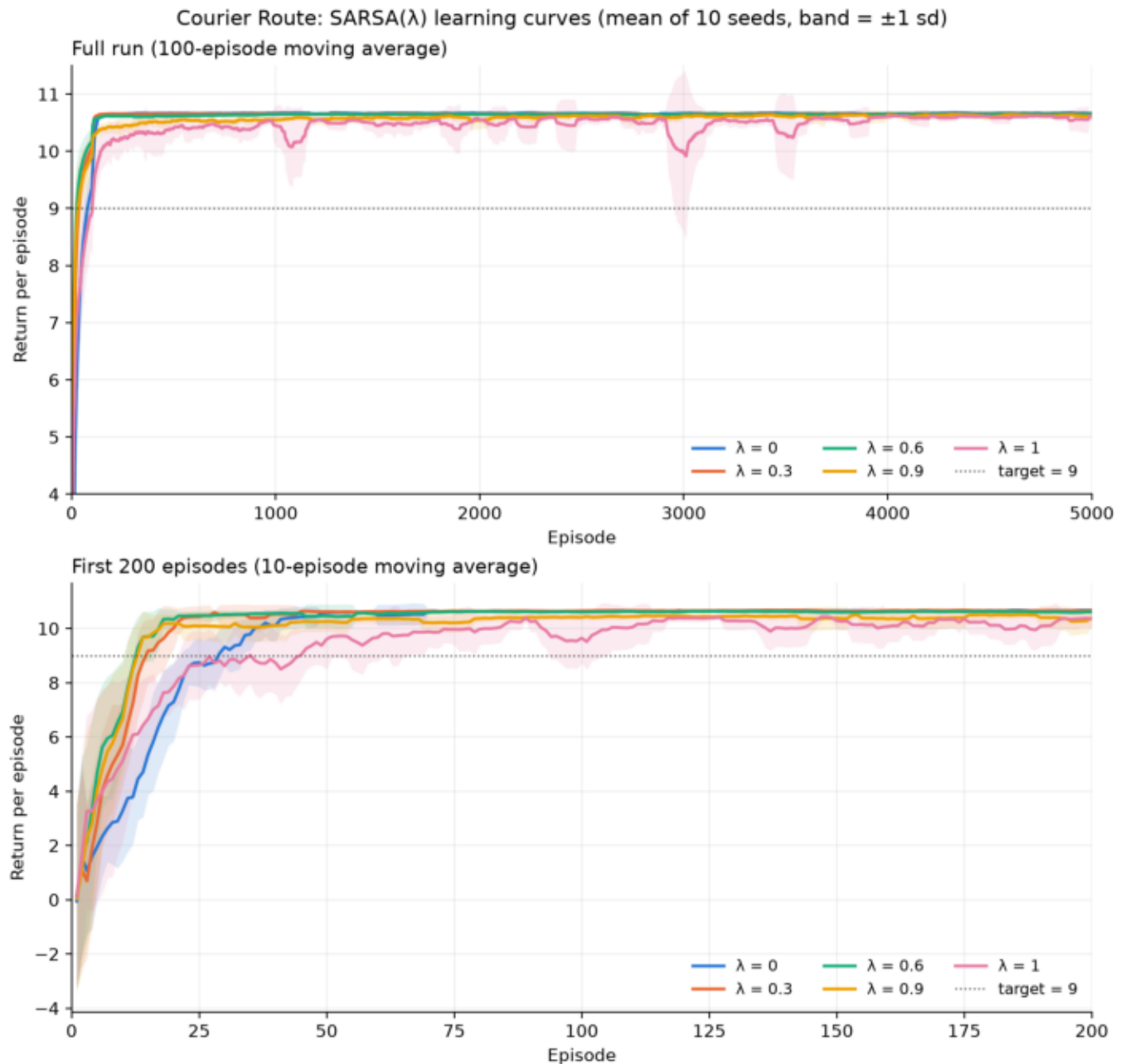


Actions 0 Up 1 Right 2 Down 3 Left
Encoding $\text{state} = \text{stage} \cdot 49 + \text{row} \cdot 7 + \text{col}$
 $\text{stage} = \text{state} // 49$
 $\text{row} = (\text{state} \% 49) // 7$
 $\text{col} = \text{state} \% 7$
Start $\text{state } 0 = (0,0)$, stage 0
Terminal $\text{state } 97 = (6,6)$, stage 1
Rewards -0.02 every step
 $+1.0$ first entry to C
 $+10.0$ entry to D after pickup
Noise 0.85 as asked, 0.075 to each perpendicular direction
Limit truncated at 100 steps
Id cs272/CourierRoute-v0

Rendered map (ansi renderer, initial state)

Courier Route
Legend: C=checkpoint, D=destination, S=start, .=open, @=courier
Stage: package not collected
@
.
.
. . . C . . .
.
.
. D
Last action: -
Last reward: +0.00

2. λ sweep: learning curves



Mean return per training episode for $\lambda \in \{0, 0.3, 0.6, 0.9, 1.0\}$, averaged over 10 seeds; the band is ± 1 standard deviation across seeds. Top: the whole 5000-episode run, smoothed with a 100-episode trailing window. Bottom: the first 200 episodes with a 10-episode window, because every λ has learned the route within a few hundred episodes and the top panel compresses that part. The dotted line is the target return 9.

3. λ sweep: results

λ	Episodes to reach 9 (mean curve)	Per seed, mean $\pm$ sd (seeds reaching it)	Final return, last 100 (mean $\pm$ sd over seeds)	Worst 100-ep. mean after episode 500
0	78	73 $\pm$ 30 (10/10)	10.665 $\pm$ 0.010	10.54
0.3	36	34 $\pm$ 17 (10/10)	10.650 $\pm$ 0.017	10.58
0.6	25	25 $\pm$ 13 (10/10)	10.647 $\pm$ 0.018	10.52
0.9	31	33 $\pm$ 19 (10/10)	10.611 $\pm$ 0.051	10.10
1	101	72 $\pm$ 45 (10/10)	10.611 $\pm$ 0.072	5.82

Threshold: the 100-episode trailing mean of the return, averaged over the 10 seeds, first reaches 9 (the best possible return without slips is 10.76). Because that window includes the early bad episodes, the first 99 points average the episodes seen so far. The per-seed column applies the same rule to each seed on its own.

Setup: seeds 11, 22, 33, 44, 55, 66, 77, 88, 99, 110; 5000 episodes per run; $\gamma=0.99$, $\alpha=0.08$, $\epsilon=0.1$ (fixed), initial $Q=1.0$, accumulating traces. Each run seeds the agent, and the agent seeds the environment once from its own generator, so running python myrunner.py regenerates every number and figure here exactly.

Supplementary: how far one TD error travels

λ	1 step back	3 steps back	5 steps back	11 steps back
0	0%	0%	0%	0%
0.3	29.7%	2.6%	0.2%	<0.1%
0.6	59.4%	21.0%	7.4%	0.3%
0.9	89.1%	70.7%	56.2%	28.1%
1	99.0%	97.0%	95.1%	89.5%

Share $(\gamma\lambda)^k$ of a TD error that reaches the state-action pair visited k steps earlier (accumulating trace, pair visited once). On a shortest route the first decision is 5 steps before the error produced by the +1 pickup and 11 steps before the one produced by the +10 delivery.

Supplementary: accumulating vs replacing traces at $\lambda = 1$

λ	Episodes to reach 9 (mean curve)	Per seed, mean $\pm$ sd (seeds reaching it)	Final return, last 100 (mean $\pm$ sd over seeds)	Worst 100-ep. mean after episode 500
1, accumulating	101	72 $\pm$ 45 (10/10)	10.611 $\pm$ 0.072	5.82
1, replacing	101	74 $\pm$ 42 (10/10)	10.621 $\pm$ 0.052	9.30

4. Why does λ change the learning curve on this environment?

The reward sits far from the decisions that earn it. On a shortest route the first decision comes 5 steps before the TD error produced by the +1 pickup and 11 steps before the one produced by the +10 delivery, and every step in between pays only -0.02 , which says nothing about whether a move was good. What matters arrives at the end of a chain of about a dozen state-action pairs.

$\lambda = 0$ (one-step SARSA) corrects only the pair that produced an error. The +10 first reaches the last pair of the route; the pair before it can learn from it only when it is next visited, and so on, so the value crawls backwards one step per visit, and each visit closes only $\alpha = 0.08$ of the remaining gap. That is why $\lambda = 0$ needs 78 episodes to reach the target (73 ± 30 per seed).

An eligibility trace lets one TD error travel back along the whole path in the same episode: the pair visited k steps earlier receives $(\gamma\lambda)^k$ of it (table on page 3). With $\lambda = 0.9$ the first decision still receives 28.1% of the delivery error and 56.2% of the pickup error; with $\lambda = 0.6$ it receives 0.3% and 7.4%; with $\lambda = 0.3$, $<0.1\%$ and 0.2%, but the nearest pairs still get a useful share (29.7% one step back). Reaching the target takes 36 episodes at $\lambda = 0.3$, 25 at $\lambda = 0.6$ and 31 at $\lambda = 0.9$, against 78 for $\lambda = 0$, so the fastest interior λ (0.6) is $3.1\times$ faster.

Beyond about 0.6 there is no further gain: $\lambda = 0.6$ and 0.9 differ by less than the seed-to-seed spread (25 ± 13 vs 33 ± 19 episodes per seed), so I do not claim one is faster. A plausible reason is that the trace only has to bridge the ≈ 6 steps between milestones: the stage in the state makes the pickup a sub-goal, the +1 gives the first half of the route its own target, and once the second half is learned the pickup state bootstraps from it. Carrying credit further than that mostly hands each error to moves that did not cause it.

$\lambda = 1$ does worst. It is slower than every λ from 0.3 to 0.9, no faster than $\lambda = 0$ (101 episodes on the mean curve against 78; 72 ± 45 against 73 ± 30 per seed), and much less stable: its worst 100-episode mean after episode 500 falls to 5.8, against 10.1 or better for every other λ , and its final return varies more across seeds (sd 0.07 against 0.01 at $\lambda = 0$). With $\gamma\lambda = 0.99$ the trace barely decays over a route this long (the first decision still receives 89.5% of the delivery error), so each update is close to a Monte-Carlo return, whose noise here comes from the 15% slips and the 10% exploratory moves, and it is spread over every pair on the way. The replacing-trace ablation (page 3) tests part of this. Replacing traces cap a pair that is revisited (common on a slippery floor) at trace 1 instead of letting it grow, and they lift the worst dip from 5.8 to 9.3, but learning is not faster (101 vs 101 episodes). So the late instability can be attributed to accumulation on revisits, whereas the slow start cannot, and is consistent with undecayed, noisy credit.

In short, λ has to be large enough to carry the reward across the ≈ 6 steps between milestones, which one-step SARSA cannot, and small enough that $\gamma\lambda$ decays before it smears the noise of the slips over the whole episode. $\lambda = 0$ is too short, $\lambda = 1$ too long, and 0.6 to 0.9 is the useful middle on this map. After the first hundred or so episodes every $\lambda < 1$ reaches the same near-optimal return (final returns 10.6 to 10.7 against 10.76 without slips), so λ changes how fast the route is learned, not how good it ends up.

5. Sample episode of the trained greedy policy (ansi renderer)

Agent trained with $\lambda = 0.6$ (the fastest λ in the sweep) for 5000 episodes, seed 123; then one episode acting greedily ($\epsilon = 0$). Return: 10.76 in 12 steps, delivery completed (the best possible return without slips is 10.76). The frames below are the renderer's output after each step of that exact episode, with the two header lines shown once:

Courier Route

Legend: C=checkpoint, D=destination, S=start, .=open, @=courier

Steps t:action(reward), with U/R/D/L = up/right/down/left: 1:R(-0.02) 2:D(-0.02) 3:R(-0.02) 4:D(-0.02) 5:D(-0.02) 6:R(+0.98) 7:R(-0.02) 8:R(-0.02) 9:R(-0.02) 10:D(-0.02) 11:D(-0.02) 12:D(+9.98)

start

```
Stage: package not collected
@ . . . . .
. . . . .
. . . . .
. . . C . . .
. . . . .
. . . . .
. . . . . D
Last action: -
Last reward: +0.00
```

t = 3

```
Stage: package not collected
S . . . . .
. . @ . . . .
. . . . .
. . . C . . .
. . . . .
. . . . .
. . . . . D
Last action: R
Last reward: -0.02
```

t = 6

```
Stage: package collected
S . . . . .
. . . . .
. . . . .
. . . @ . . .
. . . . .
. . . . .
. . . . . D
Last action: R
Last reward: +0.98
```

t = 9

```
Stage: package collected
S . . . . .
. . . . .
. . . . .
. . . . . @
. . . . .
. . . . .
. . . . . D
Last action: R
Last reward: -0.02
```

t = 12

```
Stage: package collected
S . . . . .
. . . . .
. . . . .
. . . . .
. . . . .
. . . . . @
Last action: D
Last reward: +9.98
```

t = 1

```
Stage: package not collected
S @ . . . . .
. . . . .
. . . . .
. . . C . . .
. . . . .
. . . . .
. . . . . D
Last action: R
Last reward: -0.02
```

t = 4

```
Stage: package not collected
S . . . . .
. . . . .
. . @ . . . .
. . . C . . .
. . . . .
. . . . .
. . . . . D
Last action: D
Last reward: -0.02
```

t = 7

```
Stage: package collected
S . . . . .
. . . . .
. . . . .
. . . @ . . .
. . . . .
. . . . .
. . . . . D
Last action: R
Last reward: -0.02
```

t = 10

```
Stage: package collected
S . . . . .
. . . . .
. . . . .
. . . . . @
. . . . .
. . . . .
. . . . . D
Last action: D
Last reward: -0.02
```

t = 2

```
Stage: package not collected
S . . . . .
. @ . . . . .
. . . . .
. . . C . . .
. . . . .
. . . . .
. . . . . D
Last action: D
Last reward: -0.02
```

t = 5

```
Stage: package not collected
S . . . . .
. . . . .
. . . . .
. . @ C . . .
. . . . .
. . . . .
. . . . . D
Last action: D
Last reward: -0.02
```

t = 8

```
Stage: package collected
S . . . . .
. . . . .
. . . . .
. . . @ . . .
. . . . .
. . . . .
. . . . . D
Last action: R
Last reward: -0.02
```

t = 11

```
Stage: package collected
S . . . . .
. . . . .
. . . . .
. . . . . @
. . . . .
. . . . . D
Last action: D
Last reward: -0.02
```